

VEDAT ANWAR

Mechanical Engineer

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Education

San Diego State University

Mechanical Engineering | GPA: 3.1/4.0

San Diego, United States

08/2020 - 12/2024

- Dean's List, Mechanical Engineering, 2020-2021: Achieved top 10% GPA.

Projects

Battery Module Housing and Structural Case Design

American Lithium Energy Corporation (ALE)

Vista, United States

02/2026 - Current

Designed and validated structural battery enclosure and production tooling for cylindrical cell manufacturing, integrating electrode handling, precision cutting, and mechanical assembly constraints to support repeatable high-throughput production.

- Designed mold and fixture components in SolidWorks and machined using CNC milling and lathe processes to maintain part tolerance at 0.10 mm and ensure repeatable enclosure assembly
- Integrated electrode manufacturing constraints including calendaring, slitting, and CCD inspection with positioning accuracy at 0.10 mm and defect detection threshold below 0.5%
- Validated mechanical joints using Instron tensile testing on stainless steel fasteners with load capacity exceeding 900 MPa and displacement below 0.20 mm under load
- Maintained dimensional tolerances to meet drawing and process requirements during laser slitting at 120 PPM

Dry Room Dehumidifier and Chiller System Integration

American Lithium Energy Corporation (ALE)

Vista, United States

10/2025 - 02/2026

Designed and coordinated dry room dehumidification and HVAC system using AutoCAD, integrating supply, return, and exhaust air with chilled water and filtration units to maintain low humidity and meet mechanical code requirements.

- Integrated dual dehumidifier units at 95 kW with chilled water flow at 16.34 m³/h and coordinated electrical loads at 29.1 kW and 31.1 kW
- Routed supply return and exhaust duct systems including 400×300 mm inlets and 400×400 mm columns with vertical risers at 4.0 m for airflow continuity
- Coordinated HEPA filtration at 1000 m³/h and 1500 m³/h and connected exhaust to activated carbon system for air quality compliance
- Integrated chilled water piping pumps and tanks and allocated maintenance clearance zones for installation access and serviceability

Experience

American Lithium Energy Corporation (ALE)

Mechanical Engineer

Vista, United States

09/2025 - Current

- Support R&D pilot line operations including machine setup validation testing process verification for repeatable production
- Execute validation including burn in ATE verification and testing against defined specifications acceptance criteria
- Review drawings, BOM revisions, and change documentation for traceability and compliance with internal standards
- Analyze failure modes support root cause corrective action continuous improvement feedback
- Interface engineering manufacturing customers to resolve build issues and meeting technical regulatory requirements

The Countertop Company (TCC)

CNC Programmer / Production Manager

Escondido, United States

02/2025 - 09/2025

- Introduce 5S (sort, set in order, shine, standardize, sustain) into production to improve efficiency and maintain repeatability
- Design CAD geometry from drawings apply GD&T assign blade cutting for bulk features precision cutting for unique features
- Analyze machining variation identify root causes implement corrective adjustments maintain dimensional accuracy

Skills

- CAD & Simulation:** SolidWorks, AutoCAD, Ansys (FEA), MATLAB, Structural Dynamics
- Programming & Automation:** C/C++, Data Processing, Test Automation
- Engineering & Compliance:** GD&T, QA/QC, ASTM, DO-160, NFPA, SOPs, DOE
- Prototyping:** Additive Manufacturing, water jetcutting, CNC Lathe Machining, Model Testing, Technical Reporting

Languages

- English:** Native Proficiency
- Mandarin Chinese:** Full Professional Proficiency